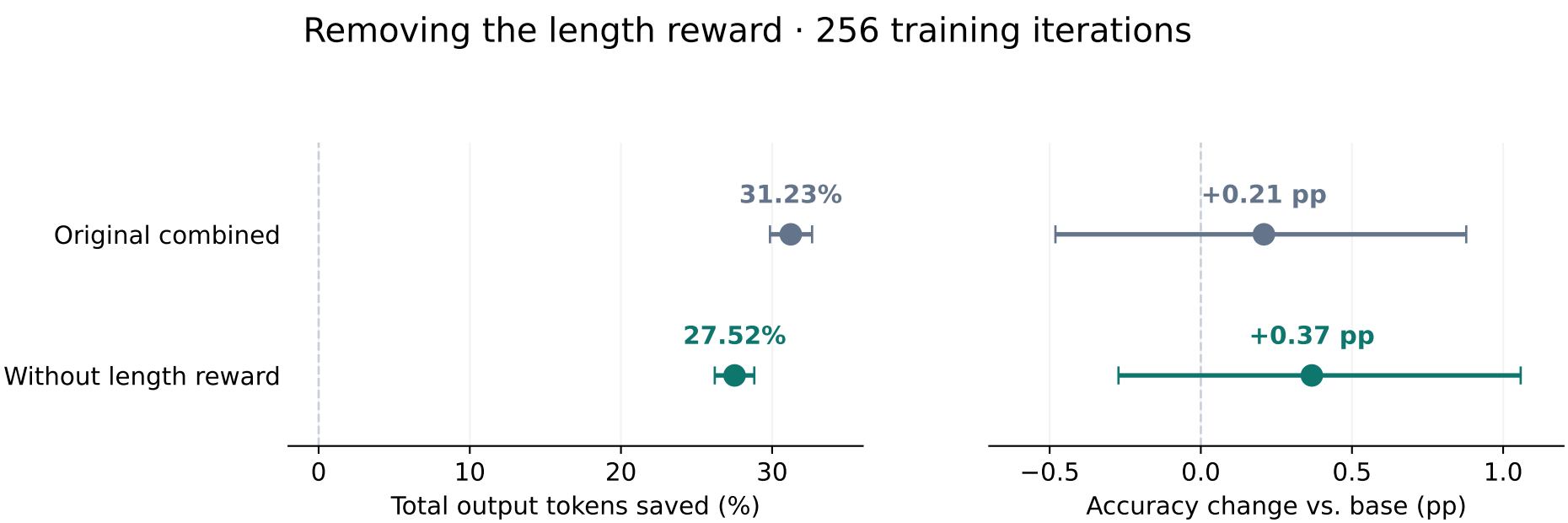
Removing the length reward · 256 training iterations
Original combined
Without length reward
31.23%
27.52%
+0.21 pp
+0.37 pp
0
10
20
30
Total output tokens saved (%)
−0.5
0.0
0.5
1.0
Accuracy change vs. base (pp)
Full evaluation: 2,759 questions × 8 responses · 95% paired-prompt intervals vs. base
One training seed; historical combined control; repeatedly inspected benchmark.
Intervals omit training-seed and selection uncertainty; no between-method significance test.